Yanze Wu

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Education

- **George Mason University** — Virginia, U.S.
  Ph.D. in Information Technology. — *Jan. 2024 – May 2027 (Expected).*
- **Hohai University** — Changzhou, China
  Master of Communication and Information Systems. — *Sep. 2020 – Jun. 2023*
- **Hohai University** — Changzhou, China
  Bachelor of Electronic Science and Technology. — *Sep. 2016 – Jun. 2020*

Awards

- **63rd Design Automation Conference (DAC) Young Fellow** — California, U.S.
  March. 2026.
- **ACM Student Research Competition** — California, U.S.
  Third-place *winner in PACT 2025 student competition .* — *Nov. 2025.*
- **Graduate Student Travel Fund** — Virginia, U.S.
  Awarded to students giving presentation and provides financial support for conference-related travel. — *Sept. 2025.*
- **Daniel R. Bannister PhD fellowship** — Virginia, U.S.
  Supported by the Bannister family. Awarded to the ***top 4*** *graduate students at GMU.* — *Dec. 2024.*
- **National Scholarship** — Changzhou, China
  National Level. Awarded to ***top 2%*** *students in the college.* — *Oct. 2022.*
- **First-class Scholarship** — Changzhou, China
  Bachlor's and Master's Level. Awarded to ***top 10%*** *students in the college.* — *2017/2018/2019/2020/2021.*

Publications

[C6] **Y. Wu**, Devin Park, Tanvir Arafin. "ATTINA: Spatial Acceleration of Additive Number Theoretic Transform for Energy-Efficient zkSNARKs at the Edge," **DAC'26**, California, U.S. **(22.3% acceptance rate)**

[C5] Andrew Fan, **Y. Wu**, Tanvir Arafin. "GPU Acceleration of the Sum-Check Protocol Over Binary Tower Fields for Verifiable Computing," **DATE'25**, Verona, Italy, doi: 10.23919/DATE69613.2026.11539722. **(25% acceptance rate)**

[C4] **Y. Wu**, Qian Wang, Tanvir Arafin. "SHAFI: Securing Hash-Based Post-Quantum Cryptography from Hardware Fault Injection Attacks," **AsianHOST'25**, Nanjing, China, doi: 10.1109/AsianHOST68425.2025.11370367.

[**C3**] **Y. Wu**, Tanvir Arafin. "Energy-Efficient Acceleration of Hash-Based Post-Quantum Cryptographic Schemes on Embedded Spatial Architectures," **PACT'25**, Irvine, California, USA, doi: 10.1109/PACT65351.2025.00033. **(27.8% acceptance rate, 34 of 122)**

[**C2**] **Y. Wu**, Tanvir Arafin. "Accelerating Torus Fully Homomorphic Encryption on Energy-Efficient Heterogeneous Architecture," **ICCD'25**, Dallas, USA, doi: 10.1109/ICCD65941.2025.00094. **(25.5% acceptance rate)**

[**C1**] **Y. Wu** and M. T. Arafin, "Ising Model Processors on a Spatial Computing Architecture," **2024 IEEE 67th International Midwest Symposium on Circuits and Systems (MWSCAS)**, Springfield, MA, USA, 2024, pp. 1383–1387, doi: 10.1109/MWSCAS60917.2024.10658926.

[**J4**] **Y. Wu** and M. T. Arafin, "ArKANe: Accelerating Kolmogorov-Arnold Networks on Reconfigurable Spatial Architectures," **IEEE Embedded Systems Letters**, 2025, doi: 10.1109/LES.2025.3567918. **(Impact Factor 1.7)**

[**J3**] Y. Cao, **Y. Wu**, L. Qin, S. Chen, and CH. Chang, "Area, Time and Energy Efficient Multicore Hardware Accelerator for Extended Merkle Signature Scheme," **IEEE Transactions on Circuits and Systems I: Regular Papers**, 2022, doi: 10.1109/TCSI.2022.3200987. **(Impact Factor 1.18)**

[**J2**] Y. Cao, **Y. Wu**, X. Lu, S. Chen, J. Ye, and CH. Chang, "An Efficient Full Hardware Implementation of Extended Merkle Signature Scheme," **IEEE Transactions on Circuits and Systems I: Regular Papers**, vol. 69, no. 2, pp. 682–693, Feb. 2022, doi: 10.1109/TCSI.2021.3115786. **(Impact Factor 1.18)**

[**J1**] W. Zheng, **Y. Wu**, et al., "Accelerating Hybrid and Compact Neural Networks Targeting Perception and Control Domains with Coarse-Grained Dataflow Reconfiguration," **Journal of Semiconductors**, vol. 41, no. 2, 2020, doi: 10.1088/1674-4926/41/2/022401. **(Impact Factor 0.23)**

Talks and Presentations

- **DAC 2026 – Oral and Poster Presentation**: Presented "ATTINA: Spatial Acceleration of Additive Number Theoretic Transform for Energy-Efficient zkSNARKs at the Edge." *California, U.S., 2026*
- **PACT 2025 – Conference Presentation**: Presented "Energy-Efficient Acceleration of Hash-Based Post-Quantum Cryptographic Schemes on Embedded Spatial Architectures." *Irvine, California, Nov. 2025*
- **Adaptive Acceleration Arrays for Cryptography**: Achieve High-Performance Cryptographic Algorithms on Spatial Computing Architectures. *George Mason University, Apr. 2024*
- **Security on Heterogeneous Architecture**: Discussed the Evolution of Heterogeneous Computing Architectures and Security Challenges in Modern Multi-Processor Systems. *George Mason University, Feb. 2024*

Experience

- **Fisilink Microelectronics Technology Co., Ltd.** Suzhou, China
  Internship of Digital Integrated Circuit Design *Jun. 2020 - Sep. 2020*
  - **XMSS Hardware Accelerators**: Learn and implement XMSS in hardware.
  - **Electronic Control Unit**: Construct fast verify modules for vehicles at the standard of 2200 times per second.

- **Fast Modular Application Accelerators**: Construct fast modular accelerators for high-performance NTT implementation.

- **University of Chinese Academy of Sciences** Shenzhen, China
  Research Assistant *Oct. 2019 - Jan. 2020*
  - **Implementation of LSTM Hardware Accelerators**: Learn Long Short-Term Memory (LSTM) neural networks and construct hardware accelerators.
  - **Accelerators for Float-point Numbers**: Build lightweight accelerators for float-point numbers utilized in neural networks.

Skills and Interests

- **Programming skills**: Verilog, C, Python (have experience building hardware accelerators for cryptographic algorithms, modular multiplications, and neural networks.) **Board**: ZYNQ, VCK190, Artix-7

- **Research interests**: Zero-Knowledge Proof, Fully Homomorphic Encryption, Hardware accelerating, spatial computing architecture, post-quantum cryptography, side-channel attack, neural network, ising model.

Reference

- **Tanvir Arafin, Ph.D.** marafin@gmu.edu
  Assistant Professor, Cyber Security Engineering Department, George Mason University

- **Zhngdao Wang, Ph.D.** zwang52@gmu.edu
  Professor, Cyber Security Engineering Department, George Mason University